

Chapter 1

Benefit-Cost Ratio Calculator Documentation

1.1 Introduction

The `bcr_calculator.py` script calculates Benefit-Cost Ratios (BCR) and Net Benefits for transmission projects. It aggregates benefits and costs from all CTCC modules and computes multiple BCR metrics to evaluate project economics from different perspectives.

1.2 Method Overview

1.2.1 Purpose

This script calculates BCR metrics by:

1. Loading scenario data from `batch_summary.csv` (robust lookup: exact, partial, prefix, last-row fallback)
2. Aggregating benefits (congestion, curtailment; line losses are always costs, never benefits)
3. Aggregating costs by category (capital, operational, energy/emissions, risk, delay) using `safe_get_numeric` and `prefer_csv_subtotal`
4. Applying optional `BCRConfig` exclusions (`no_emissions`, `no_linelosses`, `capital_only`, `no_wildfire`, etc.) when loading from YAML
5. Calculating multiple BCR metrics (system, capital, excluding risk, excluding emissions)
6. Computing net benefits; displaying formatted summary and updating CSV

1.2.2 Calculation Approach

The BCR calculation follows this structure:

$$\text{Total Benefits (Conservative)} = \text{Congestion (Haircut)} + \text{Curtailment (Haircut)} \quad (1.1)$$

$$\text{Total Costs} = \text{Capital} + \text{Operational} + \text{Energy/Emissions} + \text{Risk} + \text{Delay} \quad (1.2)$$

$$\text{BCR}_{\text{system}} = \frac{\text{Conservative Benefits}}{\text{Total Costs}} \quad (1.3)$$

$$\text{BCR}_{\text{capital}} = \frac{\text{Conservative Benefits}}{\text{Capital Costs}} \quad (1.4)$$

$$\text{BCR}_{\text{excluding_risk}} = \frac{\text{Conservative Benefits}}{\text{Total Costs} - \text{Risk Costs}} \quad (1.5)$$

$$\text{BCR}_{\text{excluding_emissions}} = \frac{\text{Conservative Benefits}}{\text{Total Costs} - \text{Emissions Costs}} \quad (1.6)$$

$$\text{Net Benefit} = \text{Conservative Benefits} - \text{Total Costs} \quad (1.7)$$

1.2.3 Inputs and Outputs

Inputs:

- Scenario ID (to load from `batch_summary.csv`)
- All cost and benefit metrics from CTCC modules

Outputs:

- BCR metrics (system, capital, excluding risk, excluding emissions) - all use conservative benefits
- Net benefits (total and excluding risk)
- Benefit and cost breakdowns
- Formatted summary display

1.2.4 Integration with Other Scripts

This script integrates with:

- `path_config.py` — `OUTPUTS_DIR`, `YAMLS_DIR` (BCR config YAML optional)
- `batch_summary.csv` — Reads scenario row by `scenario_id`; writes BCR columns back
- All CTCC cost/benefit scripts — Aggregates their outputs from the batch summary row

1.3 Key Functions

1.3.1 `load_scenario_data(scenario_id, output_dir)`

Loads scenario data from `batch_summary.csv`. Uses robust lookup: (1) exact match on `scenario_id`, (2) partial match (substring), (3) prefix match (e.g. `sample_0_...`), (4) fallback to last row (for batch sensitivity runs). Converts numeric cells via `convert_row_to_numeric`.

Parameters:

- `scenario_id` (str): Scenario identifier
- `output_dir` (str): Directory containing `batch_summary.csv` (default `OUTPUTS_DIR`)

Returns:

- Dictionary with all data for the scenario, or `None` if file missing/empty

1.3.2 `calculate_benefits(data)`

Calculates total benefits from scenario data.

Parameters:

- `data` (dict): Dictionary with scenario data

Returns:

- Dictionary with benefit breakdown and totals (PV, nominal, haircut)

Key Logic:

```
1 congestion_benefit_pv = safe_get_numeric(data, "  
    congestion_benefit_pv")  
2 curtailment_benefit_pv = safe_get_numeric(data, "  
    curtailment_benefit_pv")  
3 # Line losses are always costs, never benefits (line_loss_benefit_pv  
    = 0)  
4 calculated_total_benefits = congestion_benefit_pv +  
    curtailment_benefit_pv  
5 total_benefits_pv = prefer_csv_subtotal(data, "total_benefits_pv",  
    calculated_total_benefits)  
6 congestion_benefit_haircut = safe_get_numeric(data, "  
    congestion_benefit_haircut_pv")  
7 curtailment_benefit_haircut = safe_get_numeric(data, "  
    curtailment_benefit_haircut_pv")  
8 total_benefits_haircut_pv = prefer_csv_subtotal(data, "  
    total_benefits_haircut_pv",  
9    congestion_benefit_haircut + curtailment_benefit_haircut)
```

1.3.3 calculate_costs(data)

Calculates total costs from scenario data, organized by category.

Parameters:

- **data** (dict): Dictionary with scenario data

Returns:

- Dictionary with cost breakdown by category and totals (PV and nominal)

Cost Categories:

- **Capital:** Build, ROW, environmental mitigation
- **Operational:** O&M, insurance, line losses (greenfield), emissions
- **Risk:** Wildfire, outage
- **Delay:** Construction delay, congestion/curtailment delay, residual congestion

Key Logic:

```
1 # Capital costs
2 capital_costs_pv = build_cost_pv + row_cost_pv + env_mitigation_pv
3
4 # Operational costs (line losses only if positive - greenfield)
5 line_loss_cost_pv = max(0, line_loss_pv)
6 operational_costs_pv = oandm_pv + insurance_pv + line_loss_cost_pv +
   emissions_pv
7
8 # Risk costs
9 risk_costs_pv = wildfire_pv + outage_pv
10
11 # Delay costs
12 delay_costs_pv = (delay_cost_pv + congestion_delay_pv
13                  + curtailment_delay_pv + residual_congestion_pv)
14
15 # Totals
16 total_costs_pv = (capital_costs_pv + operational_costs_pv
17                  + risk_costs_pv + delay_costs_pv)
```

1.3.4 calculate_bcr_metrics(benefits, costs)

Calculates benefit-cost ratios and net benefits.

Parameters:

- **benefits** (dict): Dictionary with benefit breakdown
- **costs** (dict): Dictionary with cost breakdown

Returns:

- Dictionary with BCR metrics and net benefits

BCR Metrics:

- **BCR_system**: Conservative benefits / Total costs (primary metric, uses saturation-adjusted benefits)
- **BCR_capital**: Conservative benefits / Capital costs (investor perspective, uses saturation-adjusted benefits)
- **BCR_excluding_risk**: Conservative benefits / (Total costs - Risk costs) (core economics, uses saturation-adjusted benefits)
- **BCR_excluding_emissions**: Conservative benefits / (Total costs - Emissions costs) (uses saturation-adjusted benefits)

Key Logic:

```
1 # Calculate costs excluding risk
2 total_costs_excluding_risk_pv = total_costs_pv - risk_costs_pv
3
4 # Use conservative (haircut) benefits for all BCR calculations
5 total_benefits_pv = total_benefits_haircut_pv
6
7 # Calculate BCRs (with division by zero protection)
8 bcr_system = total_benefits_pv / total_costs_pv if total_costs_pv >
   0 else 0
9 bcr_capital = total_benefits_pv / capital_costs_pv if
   capital_costs_pv > 0 else 0
10 bcr_excluding_risk = (total_benefits_pv /
   total_costs_excluding_risk_pv
11                       if total_costs_excluding_risk_pv > 0 else 0)
12 bcr_excluding_emissions = (total_benefits_pv /
   total_costs_excluding_emissions_pv
13                           if total_costs_excluding_emissions_pv > 0
   else 0)
14
15 # Net benefits (using conservative benefits)
16 net_benefit_pv = total_benefits_pv - total_costs_pv
17 net_benefit_excluding_risk_pv = total_benefits_pv -
   total_costs_excluding_risk_pv
```

1.3.5 calculate_and_display_bcr(scenario_id, output_dir)

Main function to calculate and display BCR analysis.

Parameters:

- `scenario_id` (str): Unique identifier for the scenario
- `output_dir` (str): Directory containing `batch_summary.csv`

Returns:

- Dictionary with all BCR results, or None if calculation fails

1.3.6 `print_bcr_summary(benefits, costs, bcr_metrics, data)`

Prints formatted BCR summary to terminal.

1.4 Example

The following example demonstrates a typical usage:

```

1 # Example: 500 MW Overhead AC line
2 # Scenario ID: "20250101_120000"
3 #
4 # Inputs (from batch_summary.csv):
5 #   - Congestion benefit PV: $500M
6 #   - Curtailment benefit PV: $200M
7 #   - Line loss cost PV: $10M (greenfield - cost)
8 #   - Build cost PV: $100M
9 #   - ROW cost PV: $20M
10 #   - Environmental PV: $15M
11 #   - O&M PV: $50M
12 #   - Insurance PV: $5M
13 #   - Emissions PV: $8M
14 #   - Wildfire PV: $1B
15 #   - Outage PV: $500M
16 #   - Delay costs PV: $2M
17 #   - Residual congestion PV: $50M
18 #
19 # Calculation:
20 #   Total Benefits = $500M + $200M + $0 = $700M
21 #   Capital Costs = $100M + $20M + $15M = $135M
22 #   Operational Costs = $50M + $5M + $10M + $8M = $73M
23 #   Risk Costs = $1B + $500M = $1.5B
24 #   Delay Costs = $2M + $50M = $52M
25 #   Total Costs = $135M + $73M + $1.5B + $52M = $1.76B
26 #
27 #   BCR_system = $700M / $1.76B = 0.40
28 #   BCR_capital = $700M / $135M = 5.19
29 #   BCR_excluding_risk = $700M / ($1.76B - $1.5B) = 2.69
30 #   Net Benefit = $700M - $1.76B = -$1.06B
31 #
32 # Output:

```

```

33 # - BCR_system: 0.40
34 # - BCR_capital: 5.19
35 # - BCR_excluding_risk: 2.69
36 # - Net benefit: -$1.06B

```

Integration Example:

```

1 # When called from ctcc.py:
2 from scripts.bcr_calculator import calculate_and_display_bcr
3
4 # After all other scripts complete
5 scenario_id = os.environ.get('CTCC_SCENARIO_ID')
6 results = calculate_and_display_bcr(scenario_id)
7
8 # Update batch_summary.csv with BCR metrics
9 csv_manager = CTCCOutputManager()
10 csv_manager.add_bcr_metrics(results)
11 csv_manager.write_batch_summary()

```

1.5 Key Implementation Details

1.5.1 Line Loss Treatment

The script handles line losses differently for greenfield vs. reconductoring:

- **Greenfield:** Line losses are positive (costs) - counted in operational costs
- **Reconductoring:** Line losses are negative (benefits) - counted in benefits

1.5.2 Multiple BCR Perspectives

The script calculates multiple BCR metrics to provide different perspectives. All BCR metrics use conservative (saturation-adjusted) benefits to account for diminishing marginal returns:

- **System BCR:** Overall project economics including all costs (uses conservative benefits)
- **Capital BCR:** Benefits relative to capital investment (investor view, uses conservative benefits)
- **Excluding Risk BCR:** Core economics without wildfire/outage risk costs (uses conservative benefits)
- **Excluding Emissions BCR:** Economics without emissions costs (uses conservative benefits)

1.5.3 Net Benefits

The script calculates net benefits in addition to BCRs:

- **Net Benefit:** Total benefits - Total costs
- **Net Benefit Excluding Risk:** Total benefits - (Total costs - Risk costs)

1.5.4 Division by Zero Protection

All BCR calculations include checks to prevent division by zero, returning 0 if the denominator is zero or negative.

1.5.5 Data Loading

The script loads data from `batch_summary.csv` using the scenario ID, automatically converting numeric strings to floats for calculations.